



Frequency Penalty

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1) What it is

Frequency penalty discourages a model from reusing the same tokens by **lowering the probability of tokens in proportion to how often they've already appeared** (in the prompt and/or in the generated text so far).

- Result: **less repetition, more varied wording**.
 - Contrast: **Presence penalty** applies a flat hit once a token appears at least once; **frequency** scales with *how many times* it has appeared.
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2) How it works (intuition)

At each step, the model scores candidate tokens. Frequency penalty adjusts those scores **before** sampling knobs (temperature/top-p/top-k) kick in:

- **Heuristic:** `adjusted_logit = logit -  $\alpha$  * count(token)`
 - `$\alpha$` = frequency penalty strength (e.g., 0.1–1.0)
 - `count(token)` = how many times that token occurred so far
- The more a token repeats, the more it's pushed down the list.

Mental model: A progressive tax on repetition. The first reuse is cheap, the tenth is expensive.

3) When to use (and when to avoid)

Great fits

- **Long-form prose, marketing copy, brainstorming** – Avoids echoing phrases/sentences.
- **Lists** – Encourages different openings and vocabulary across bullets.
- **High-temperature generations** – Reduces “looping” or monotonous phrasing.

Use carefully / avoid

- **Code, configs, schemas** – Repetition is structural (brackets, keys); penalties can **break correctness**.
- **Legal/medical/translation** – Exact terms of art must repeat consistently.

4) Practical defaults & tuning

Task Type	Frequency Penalty	Notes
Factual Q&A, summaries	0.0–0.3	Keep wording tight; rely on structure for concision.
Explanations, reports	0.2–0.5	Smooth variety without drifting terminology.
Creative/marketing/ideas	0.4–0.8	Pushes novelty; pair with moderate temperature.
Code / JSON / strict terms	0.0	Do not penalize required tokens.

Tips

- Start low, raise by **0.1–0.2** increments.
- If synonyms get **too weird**, dial it back or lower temperature.
- Some providers support **negative values** to *encourage* repetition (rarely needed).

5) Interactions with other controls

- **Temperature / top-p / top-k**: Frequency penalty reshapes scores *before* they add randomness/diversity. High temperature + high penalty ⇒ quirky synonyms—tune gradually.

- **Presence penalty:** Use presence for “don’t repeat at all,” frequency for “if you repeat, don’t overdo it.”
- **Max tokens / stop sequences:** Penalties don’t control length; use `max_tokens` and `stop` for boundaries.

6) Prompt patterns that pair well

Varied list items

Write 8 bullets. Vary opening verbs and sentence length. Avoid repeating the same phrases across bullets.

Terminology guardrail (when penalties are low/zero)

Use the exact term “payment authorization hold” wherever applicable. Do not paraphrase this term.

Style reminder

Aim for lexical variety; avoid repeating adjectives or clause structures in consecutive sentences.

7) Troubleshooting

Symptom	Likely Cause	Fix
Awkward synonym swaps	Penalty too high (and/or temp too high)	Lower frequency penalty; reduce temperature
Still repetitive at high creativity	Penalty too low	Raise frequency penalty to 0.4–0.6
Broken code / malformed JSON	Penalizing structural tokens	Set penalty to 0; enforce structure via schema/JSON mode
Key term drifts (legal/medical)	Penalty discourages exact term	Set penalty near 0; add explicit terminology constraints

8) Copy-paste presets

Balanced prose

```
temperature=0.6, top_p=0.9, frequency_penalty=0.4
```

Idea generation / marketing

```
temperature=0.8, top_p=0.95, frequency_penalty=0.6
```

Code / structured output

```
temperature=0.2, top_p=0.6, frequency_penalty=0.0
```

9) Key takeaways

- Frequency penalty is a **scaling deterrent** against overused tokens—great for variety, risky for domains that require repetition.
- Tune **lightly** and **combine with clear prompt instructions** about style and structure.
- Use **presence penalty** when you want a simple “seen-once ⇒ discourage,” and **frequency penalty** when you need a graduated response to repetition.